

RECURSIVE DEFINITIONS AND INDUCTION

Subtitle

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OUTLINE

1. RECURSIVE DEFINITIONS

An object defined in terms of itself is said to be recursively defined. We will cover recursive definitions of the following objects:

- Functions
- Sequences

We use two steps to define a function with the set of nonnegative integers as its domain:

- **BASIS STEP**
- **RECURSIVE STEP**
- **RESTRICTION** Nothing is in the sequence other than the items defined above.

Recursively defined functions

Two steps for recursively defined functions with the set of nonnegative integers as its domain.

- BASIS STEP: Specify the value of the function at zero.
- RECURSIVE STEP: Give a rule for finding its value at an integer from its values at smaller integers.

Such a definition is called a **recursive or inductive definition**.

Example:

Suppose that f is defined recursively by,

BASIS STEP: $f(0) = 3$,

RECURSIVE STEP: $f(n + 1) = 2f(n) + 3$

Find $f(1)$, $f(2)$, $f(3)$, $f(4)$

Sol:

$$f(1) = 2f(0) + 3 = 2 * 3 + 3 = 9$$

$$f(2) = 2f(1) + 3 = 2 * 9 + 3 = 21$$

$$f(3) = 2f(2) + 3 = 2 * 21 + 3 = 45$$

$$f(4) = 2f(3) + 3 = 2 * 45 + 3 = 93$$

Example - Fibonacci Series

The Fibonacci numbers can be defined recursively as follows:

BASIS STEP: $f(0) = 0, f(1) = 1,$

RECURSIVE STEP: $f(n) = f(n - 1) + f(n - 2)$

Find $f(1), f(2), f(3), f(4)$ using the definition above.

Example - Factorial Function

The recursive definition of the factorial function:

BASIS STEP: $fact(0) = 1,$

RECURSIVE STEP: $fact(n) = n * f(n - 1)$

Find $fact(4)$

$$fact(4) = 4 * fact(3) = 4 * 3 * fact(2) = 4 * 3 * 2 * fact(1) = 4 * 3 * 2 * 1 * fact(0) = 24.$$

Example

Give a recursive definition of a^n where a is a nonzero real number and n is a nonnegative integer.

BASIS STEP: The recursive definition contains two parts. First a^0 is specified, namely,

$$a^0 = 1.$$

RECURSIVE STEP: Then the rule for finding a^{n+1} from a^n , namely $a^{n+1} = a \cdot a^n$ for $n = 0, 1, 2, 3$, is given.

These two equations uniquely define a^n for all nonnegative integers n .

For the examples that follow, there are many possible correct answers. We supply relatively simple ones.

Example

Give a recursive definition of 2^n where a is a nonzero real number and $n = 2, 3, 4, \dots$.

Sol

We write the term a_{n+1} and compare it to a_n (We can also write the term a_{n-1}).

$$\begin{aligned}a_{n+1} &= 2^{n+1} \\ &= 2 \cdot 2^n \\ &= 2 \cdot a_n\end{aligned}$$

Then check if you get the same values for $a_{n+1} = 2^n$ and the recursive definition $a_n = 2a_n$.

Example

Give a recursive definition of n^2 where $n = 1, 2, 3, \dots$.

Sol

We write the term a_{n+1} and compare it to a_n .

$$\begin{aligned}a_{n+1} &= (n+1)^2 \\&= n^2 + 2n + 1 \\&= n^2 + 1 + 2n \\&= a_n + 2n\end{aligned}$$

ASIDE: You know your answer is correct if you get the same values for $a_n = n^2$ and the recursive definition $a_{n+1} = a_n + 2n + 1$.

Example

Give a recursive definition of $a_n = 4n - 2$ where a is a nonzero real number and $n = 1, 2, 3, \dots$.

Sol

We write the term a_{n+1} and compare it to a_n .

$$\begin{aligned}a_{n+1} &= 4(n+1) - 2 \\&= 4n + 4 - 2 \\&= 4n - 2 + 4 \\&= a_n + 4\end{aligned}$$

Then check if you get the same values for $a_n = 4n - 2$ and the recursive definition $a_{n+1} = a_n + 4$.

The End